# Data & AI

11. Inside ResNet18

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# Last Time

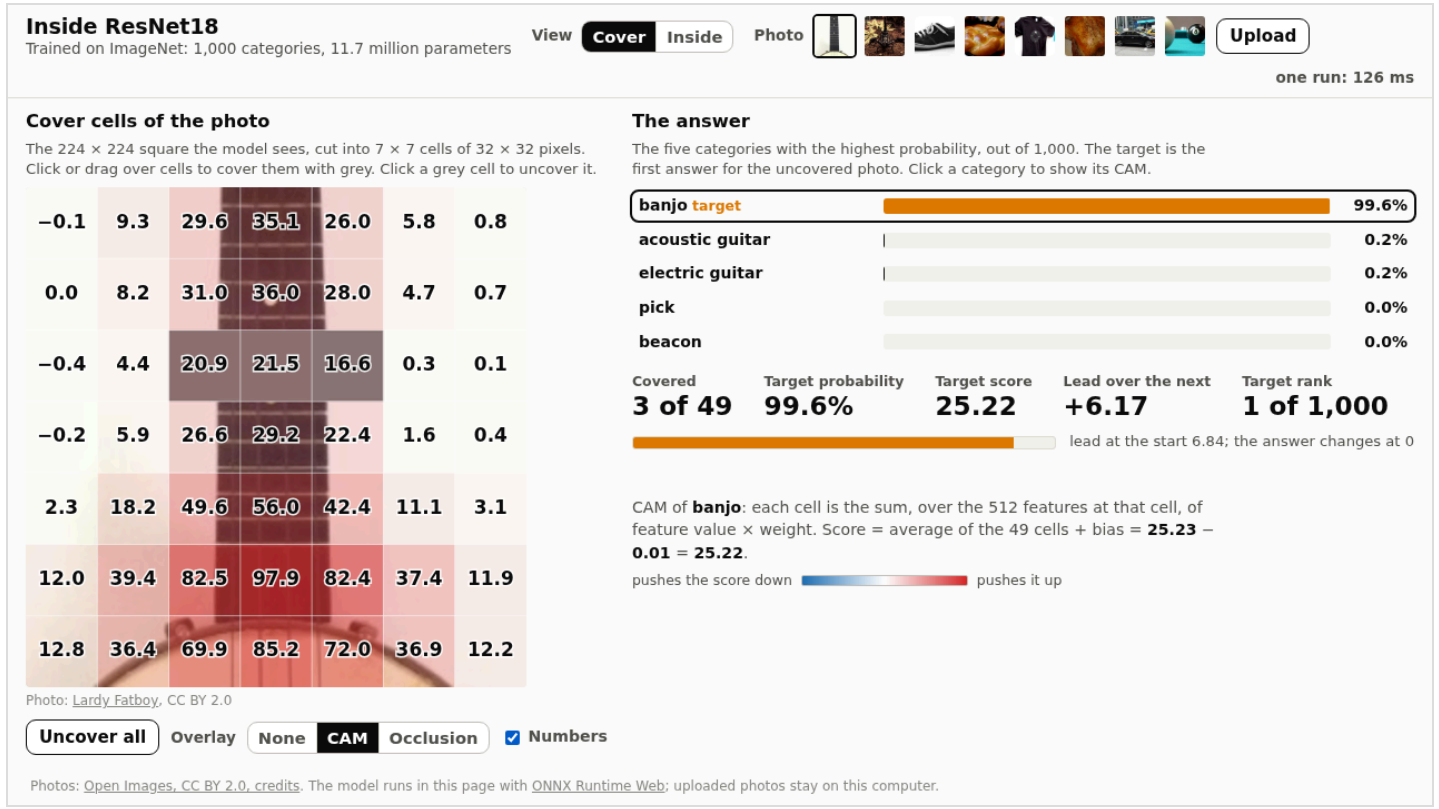
- A small CNN: convolution, max pooling, then $784 \times 10$
- ResNet18: 20 convolution layers, 11,689,512 parameters

# Today

- Inside ResNet18: what changes its answer, and why
- Gradient ascent on the pixels: feature visualization, DeepDream, style transfer
- Adversarial examples

# Inside ResNet18

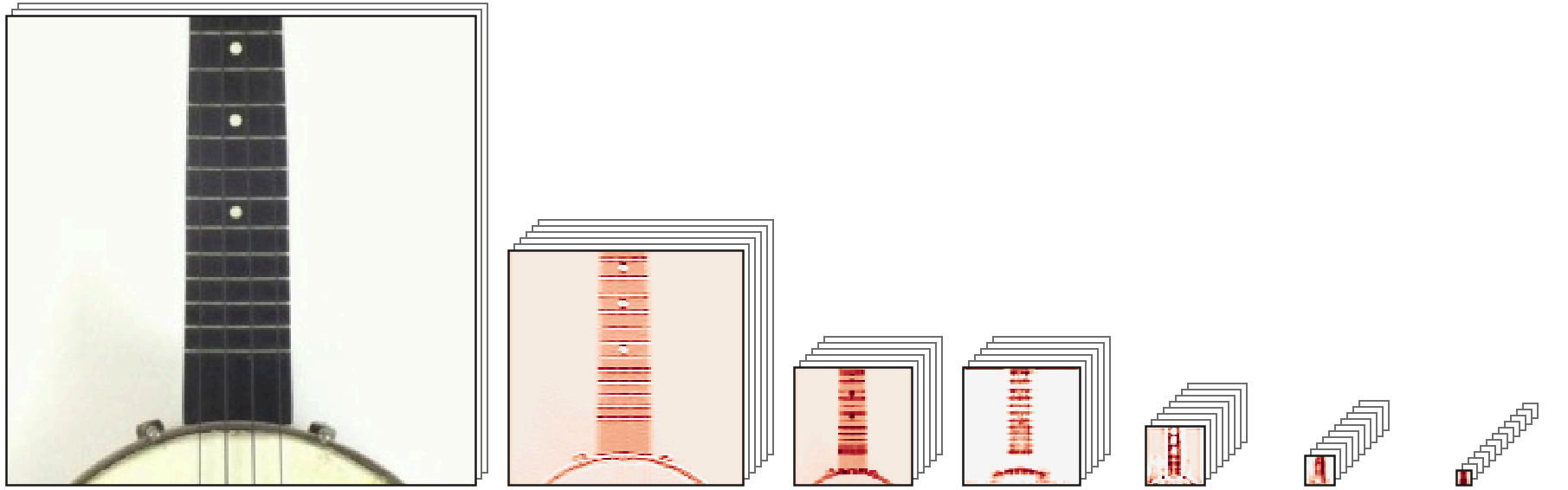
jdasam.github.io/mas1004/demos/lecture11-resnet18-cam



# The Structure of ResNet18

# ResNet18 Layer by Layer

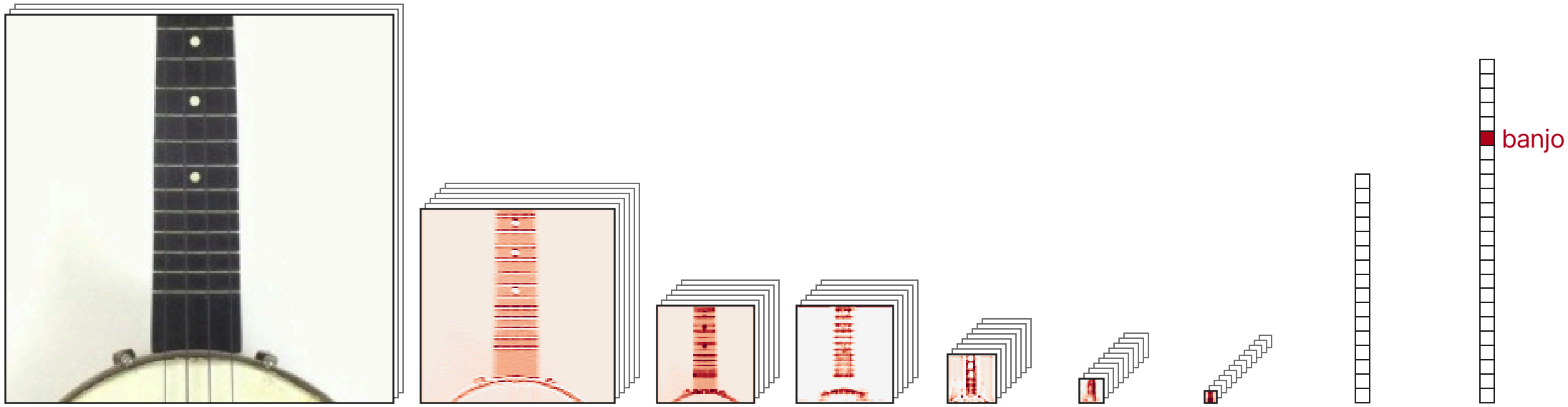
Output shape after each stage, for one 224 × 224 photo



| input | conv1 | max pool | layer1 | layer2 | layer3 | layer4 |
|---|---|---|---|---|---|---|
| 224 × 224 | 112 × 112 | 56 × 56 | 56 × 56 | 28 × 28 | 14 × 14 | 7 × 7 |
| 3 colours | 64 maps | 64 maps | 64 maps | 128 maps | 256 maps | 512 maps |

# ResNet18 Layer by Layer

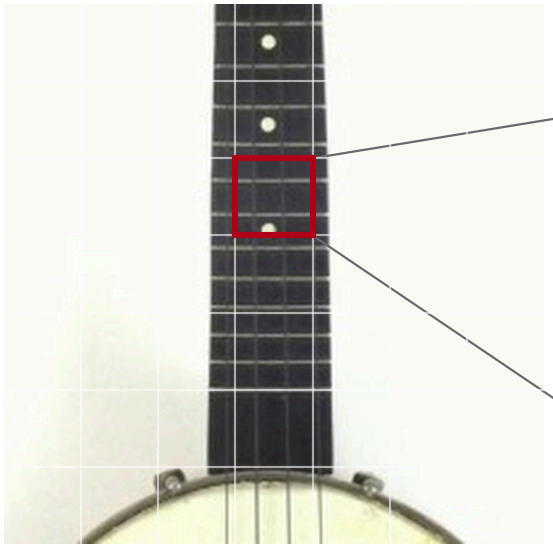
Output shape after each stage, for one 224 × 224 photo



# Receptive Field

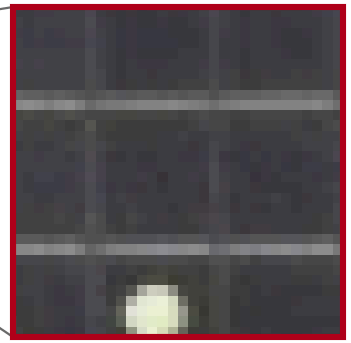
The pixels that can change one cell of layer4

7 × 7 cells



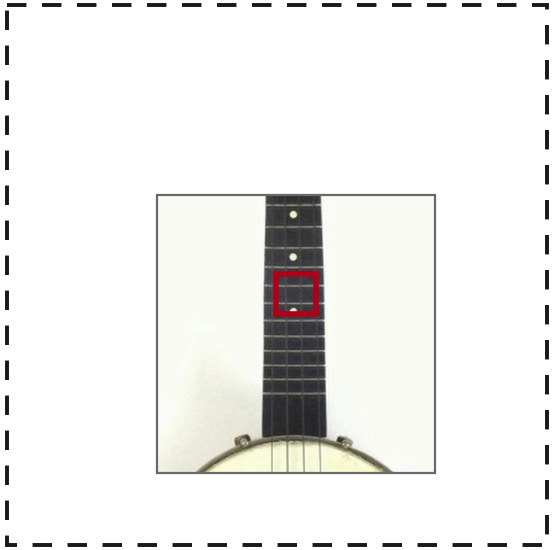
32 × 32 pixels each

row 3, column 4



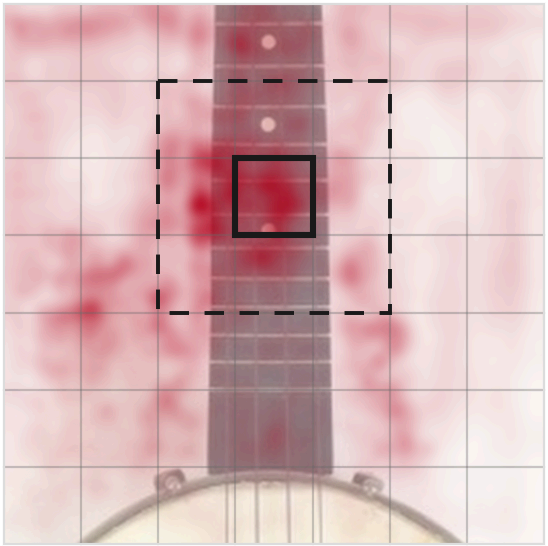
its 32 × 32 pixels

pixels that can change it



435 × 435 pixels,
more than the photo (224 × 224)

how much each pixel changes it



6% inside the cell,
33% inside the 3 × 3 cells

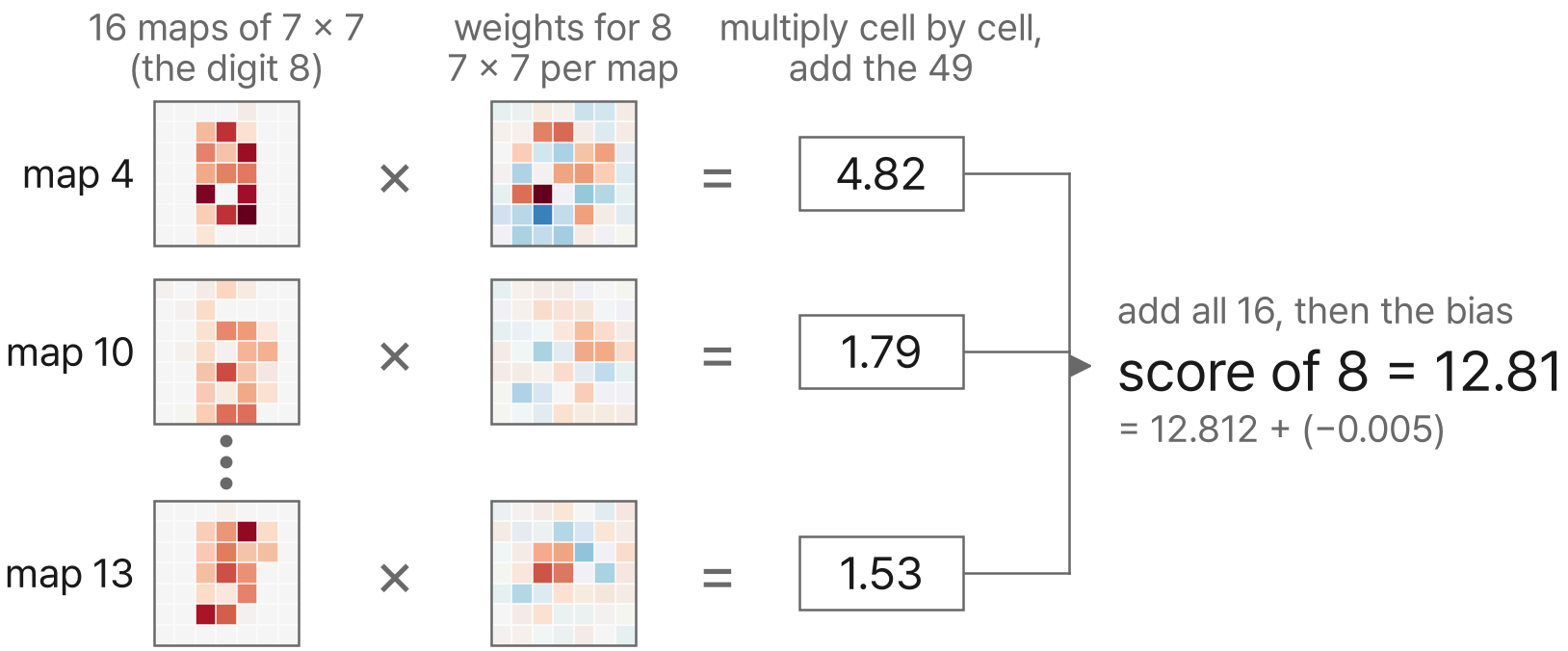
# Residual Block

He, Zhang, Ren, Sun (2016), Deep Residual Learning for Image Recognition, CVPR

- Output = F(x) + x: two convolutions compute F(x), and the input x is added back
- ResNet18: 8 blocks of two 3 × 3 convolutions

# The Last Layer of the Small CNN

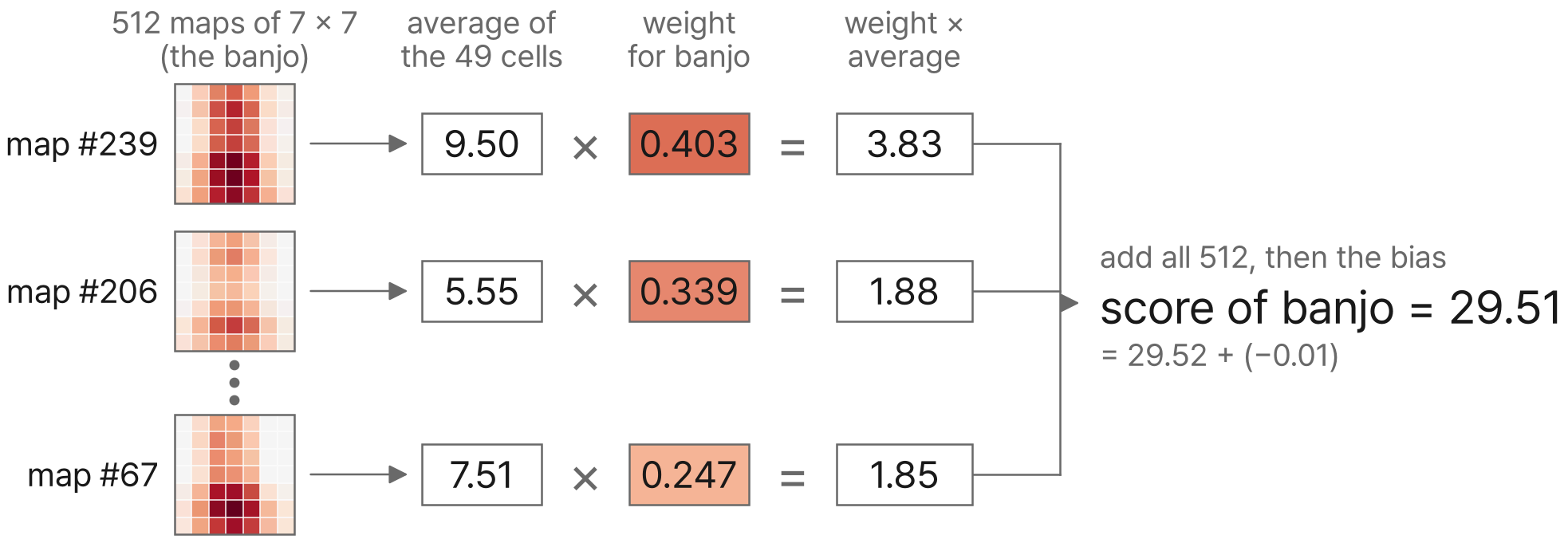
Lecture 9: 16 maps of 7 × 7, flattened into 784 numbers, then 784 × 10




Flattened: 16 × 7 × 7 = 784 numbers, each with its own weight
784 weights × 10 digits = 7,840 weights

# Global Average Pooling

ResNet18: each 7 × 7 map averaged into one number, then 512 × 1,000




Averaged: 512 maps become 512 numbers, one weight each (the same for all 49 cells)
512 weights × 1,000 categories = 512,000 weights

# Splitting the Score by Cell

$f_k(c)$: value of feature $k$ at cell $c$ · $w_k$: its weight for the category

$$\text{score} = \sum_{k=1}^{512} w_k \cdot \frac{1}{49} \sum_{c=1}^{49} f_k(c) \; + \; b$$

# Splitting the Score by Cell

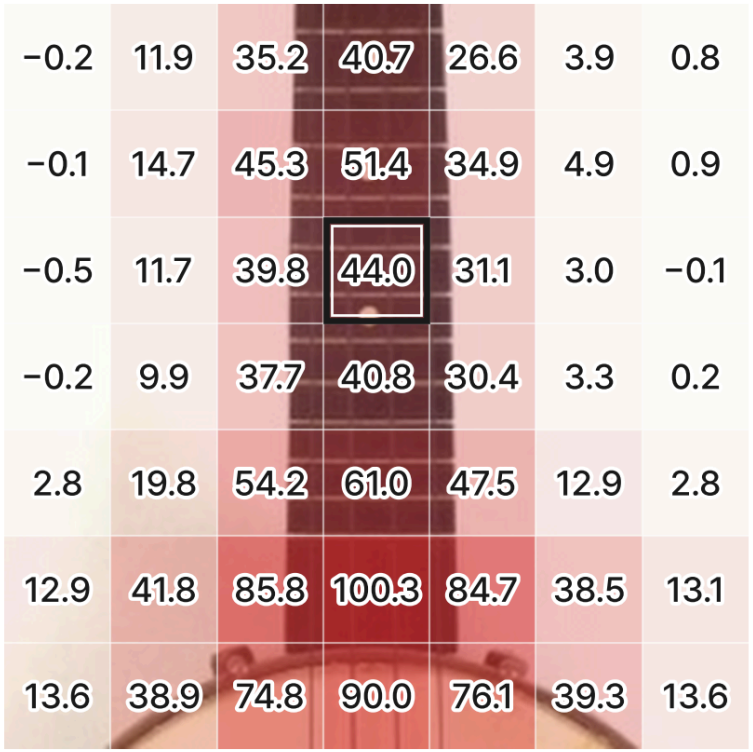
$f_k(c)$: value of feature $k$ at cell $c$ · $w_k$: its weight for the category

$$\text{score} = \sum_{k=1}^{512} w_k \cdot \frac{1}{49} \sum_{c=1}^{49} f_k(c) \; + \; b$$

$$= \frac{1}{49} \sum_{c=1}^{49} \underbrace{\sum_{k=1}^{512} w_k f_k(c)}_{\text{value of cell } c} \; + \; b$$

# Class Activation Map

Zhou, Khosla, Lapedriza, Oliva, Torralba (2016), Learning Deep Features for Discriminative Localization, CVPR



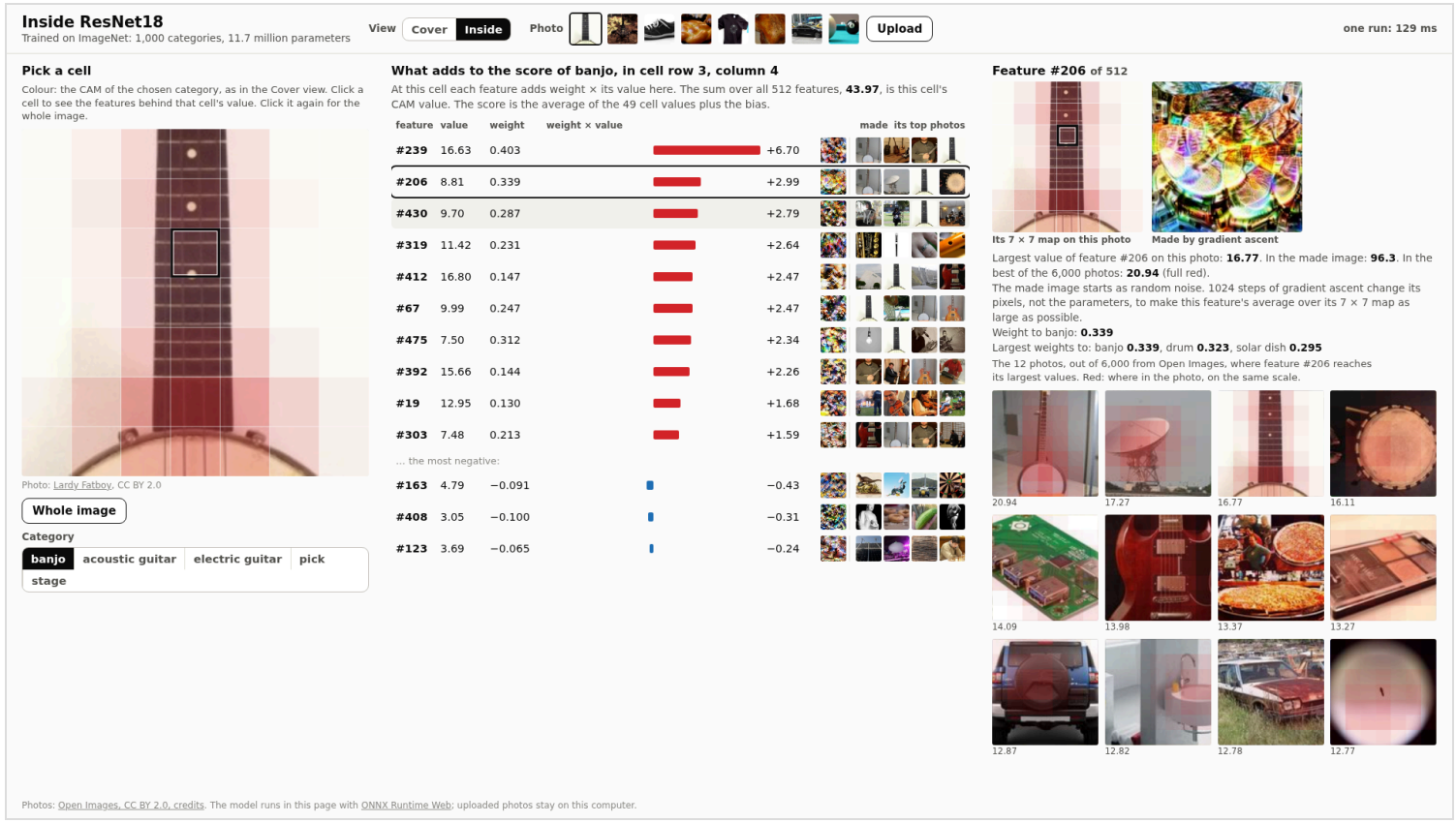

In each cell: weight × value, added over the 512 features

| | |
|---|---|
| average of the 49 cells | 29.52 |
| bias | −0.01 |
| score of banjo | 29.51 |

pushes the score down — pushes it up

# Splitting a Cell by Feature

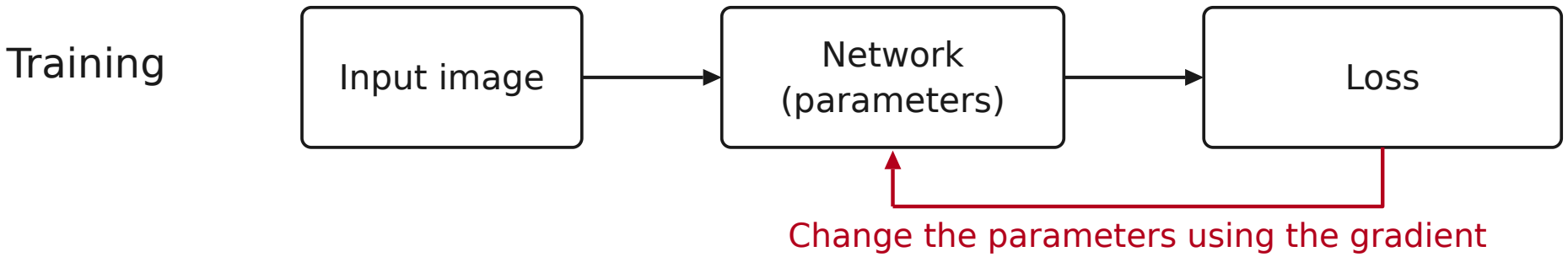
## Inside view: banjo, cell at row 3, column 4



# Gradient Ascent on the Pixels

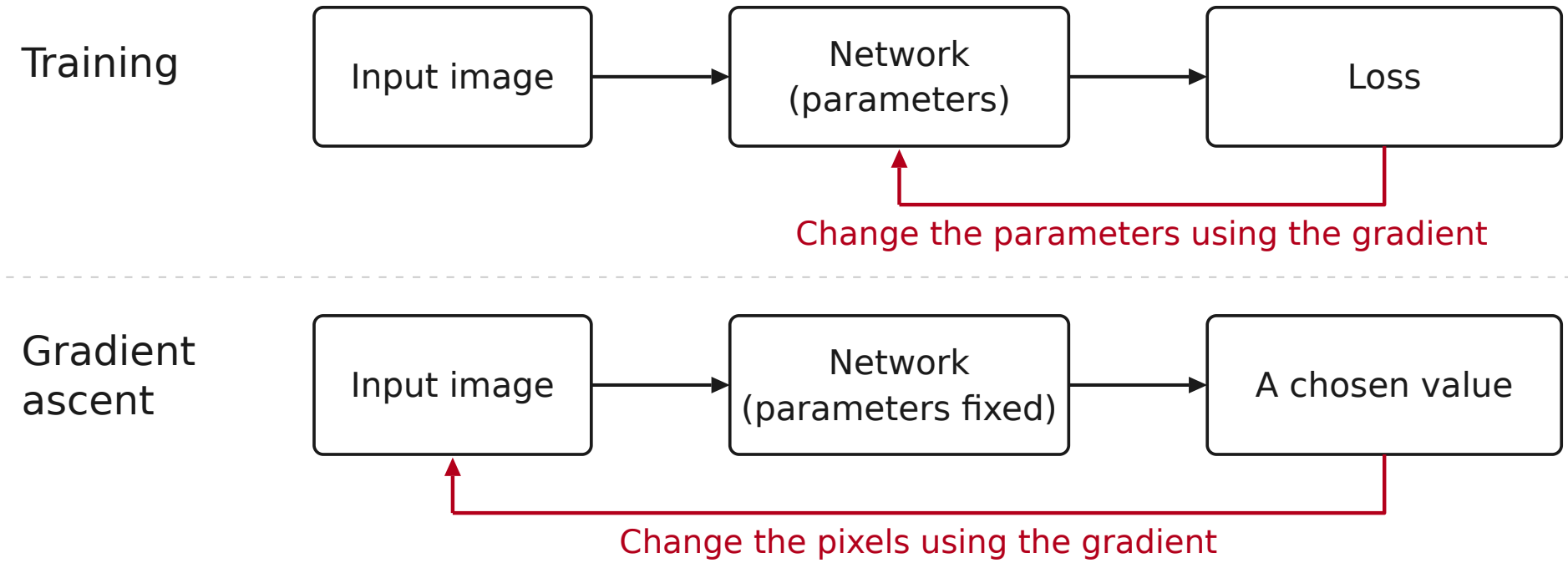
# Gradients for the Pixels

Backprop gives a gradient for every input number too



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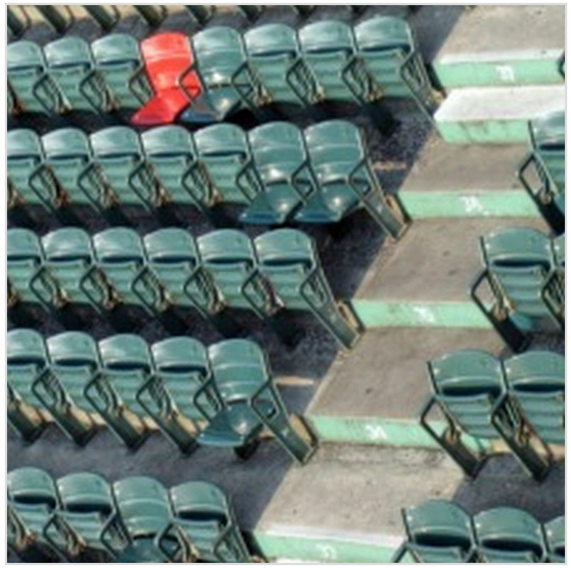


# Feature Visualization

Olah, Mordvintsev, Schubert (2017), Feature Visualization, Distill

## Best of 6,000 photos

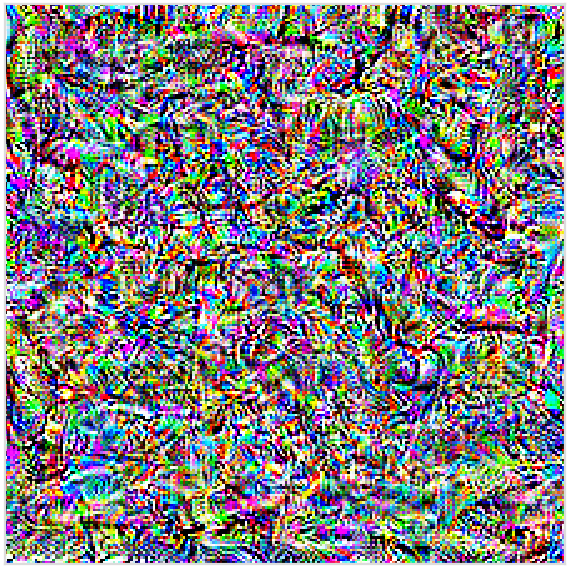
the photo where this feature is largest



feature #194: 10.4

## Plain gradient ascent

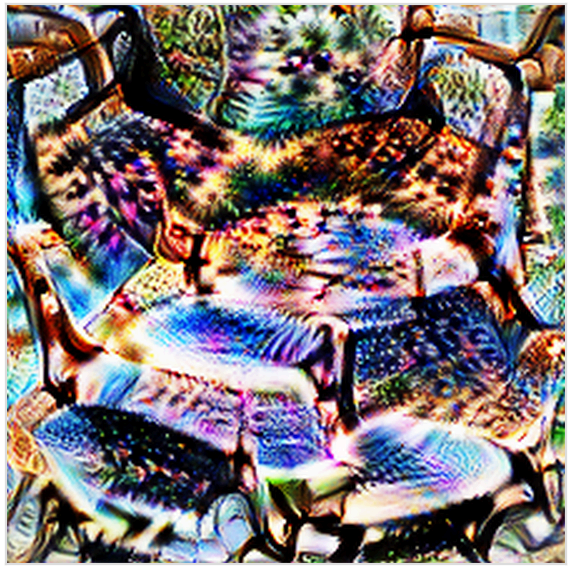
start: random pixels, 1,024 steps



feature #194: 195.9

## Regularized gradient ascent

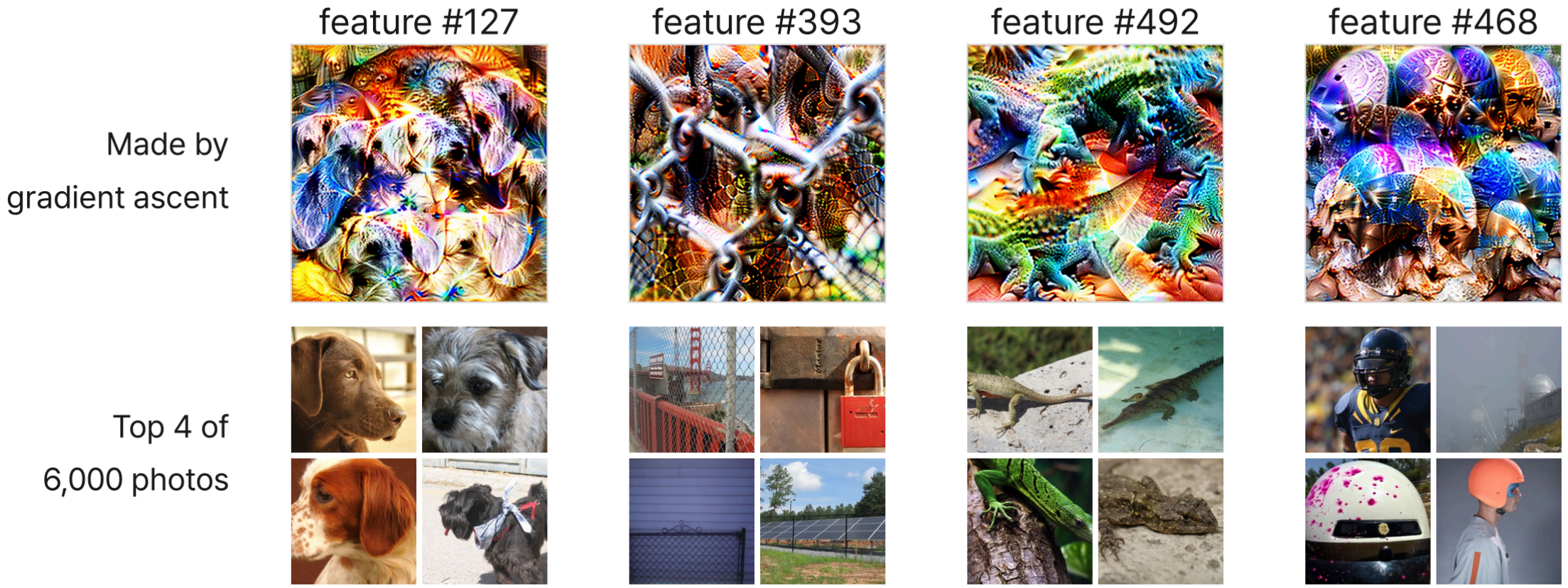
smooth pixels, random shift, scale, rotation



feature #194: 71.0

# Features and Their Top Photos

Made by gradient ascent, and the 4 photos out of 6,000 where the feature is largest



# Class Visualization

Mordvintsev, Olah, Tyka (2015), Inceptionism: Going Deeper into Neural Networks, Google Research Blog



# Class Visualization with ResNet18

Gradient ascent on the score of one category

dumbbell



score 142
probability 0.99

banjo



score 200
probability 1.00

pool table



score 142
probability 1.00

cab



score 108
probability 1.00

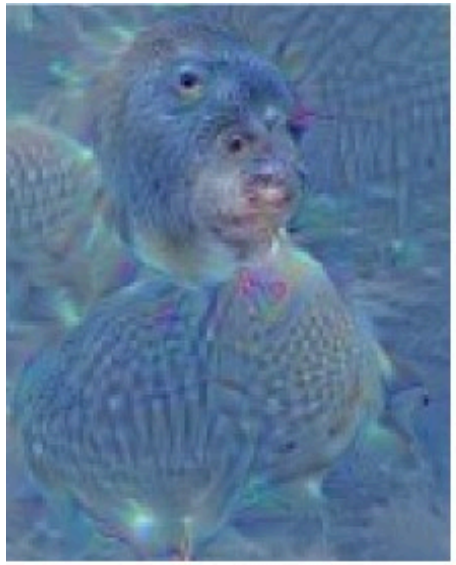
zebra



score 114
probability 1.00

# DeepDream

Mordvintsev, Olah, Tyka (2015), Inceptionism: Going Deeper into Neural Networks, Google Research Blog



"Admiral Dog!"



"The Pig-Snail"



"The Camel-Bird"



"The Dog-Fish"

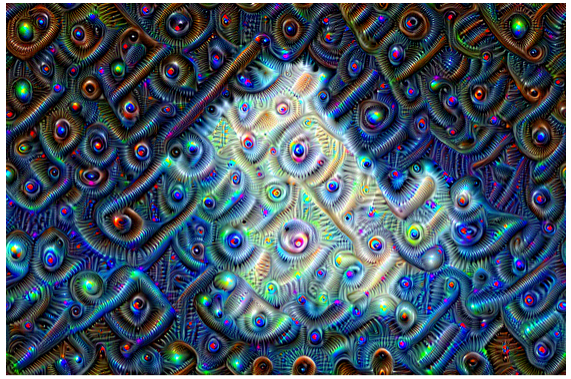
- Start from a photo, and make the values of a whole layer larger

# DeepDream by Layer

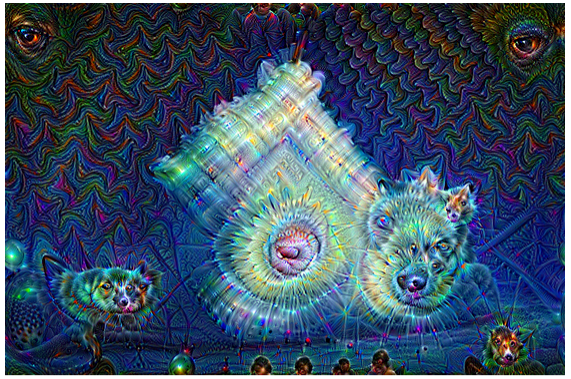
ResNet18, the same photo



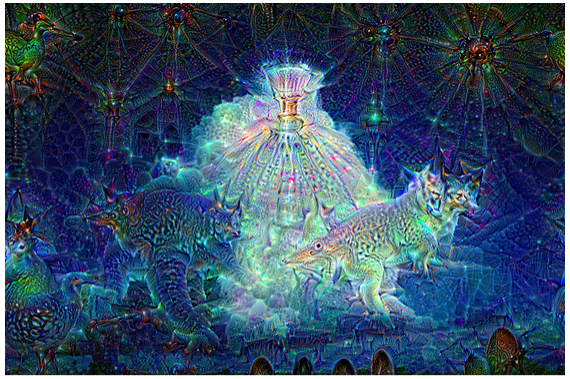
Input photo



layer2 (shallow)



layer3 (middle)

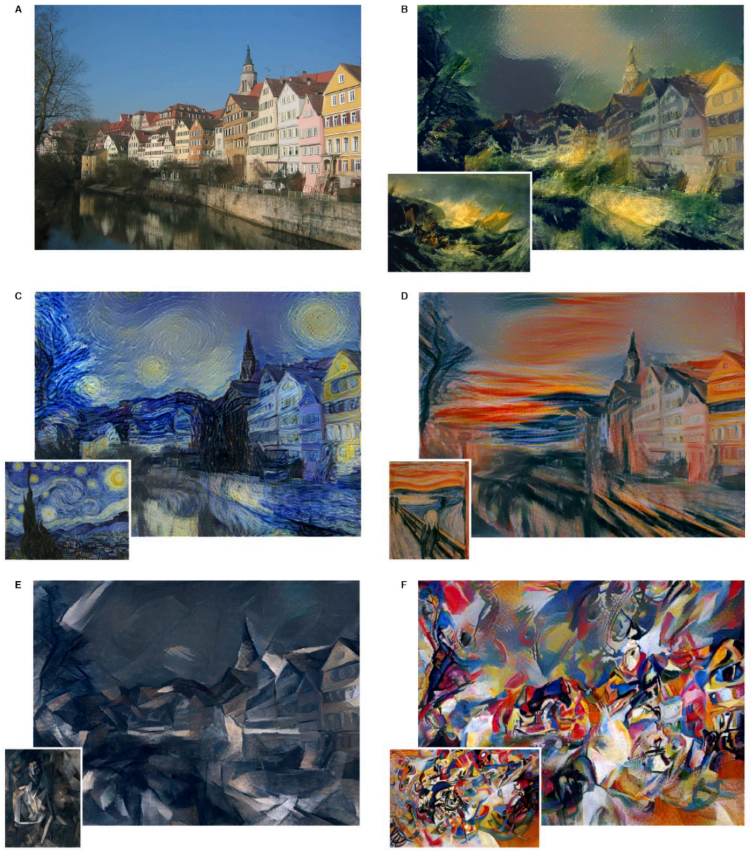


layer4, first block (deep)

Input: Ralph F. Kresge, Cumulus6 (NOAA, public domain)

# Neural Style Transfer

Gatys, Ecker, Bethge (2016), Image Style Transfer Using Convolutional Neural Networks, CVPR




- Content loss: the maps of a deep layer, cell by cell
- Style loss: which features appear together, position thrown away
- Change the pixels to lower the sum of the two losses

# Style Transfer Result

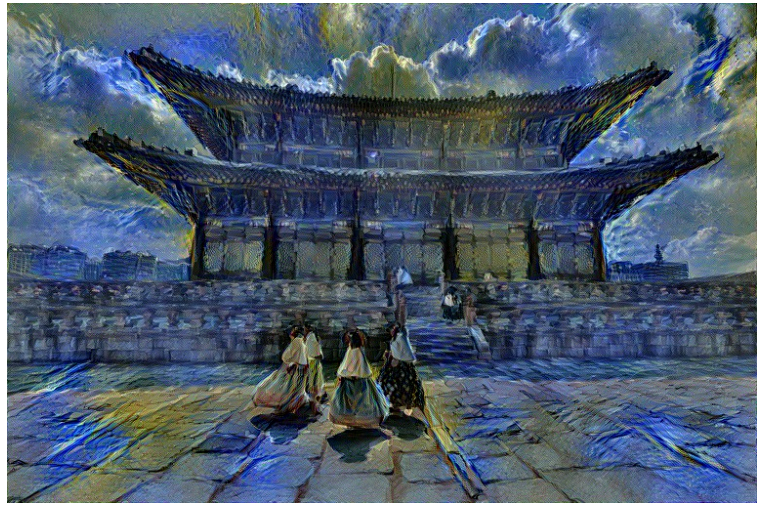
VGG19 · Gyeongbokgung photo and The Starry Night



Content



Style



Result after 400 steps from noise

Content: chsyang, Gyeongbokgung Geunjeongjeon (Public Domain Mark) · Style: Vincent van Gogh, The Starry Night

# Adversarial Examples

# Adversarial Examples

Goodfellow, Shlens, Szegedy (2015), Explaining and Harnessing Adversarial Examples, ICLR

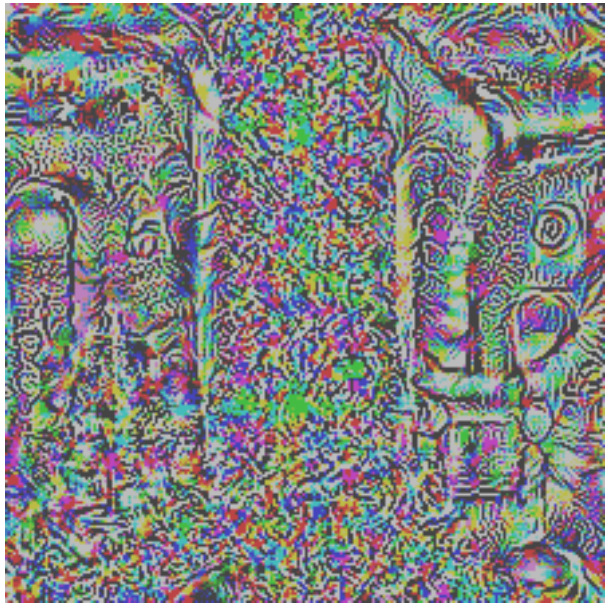
- The same gradient for the pixels, used to change the answer
- Every pixel moves by a tiny amount that a person cannot see

# An Adversarial Banjo

ResNet18, the banjo from the demo, each pixel changed by at most 1 level out of 255



banjo 99.8%



Change × 60

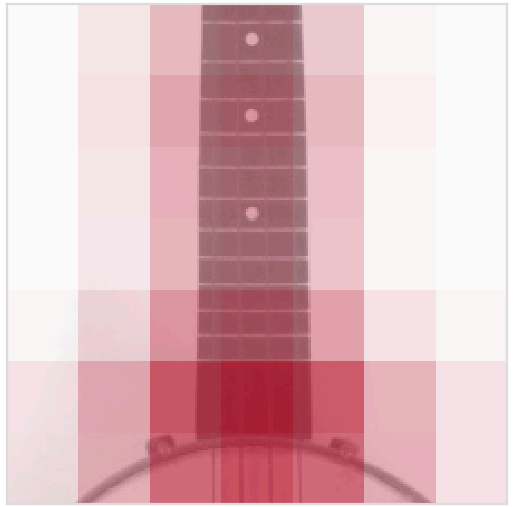


toaster 100%

# CAM of the Adversarial Banjo

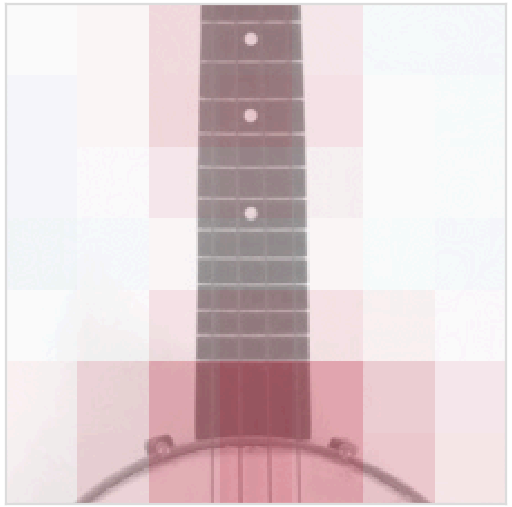
The same model, before and after the change

banjo, original photo



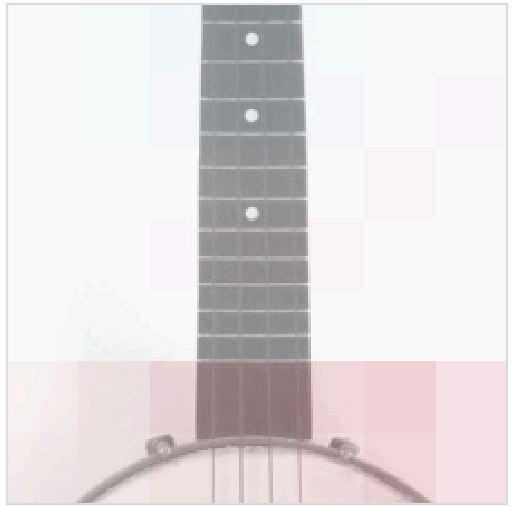
score 29.5

banjo, changed photo



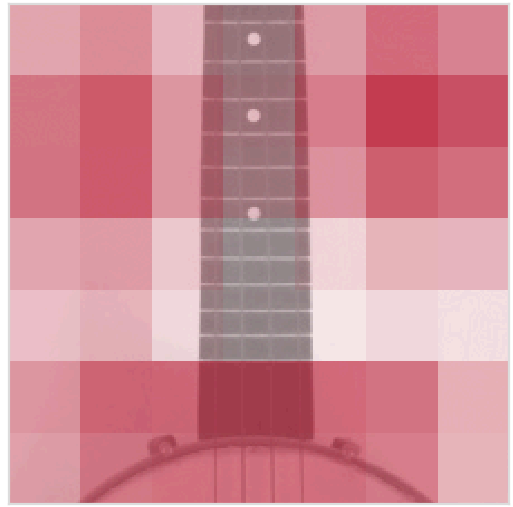
score 11.4

toaster, original photo



score 2.0

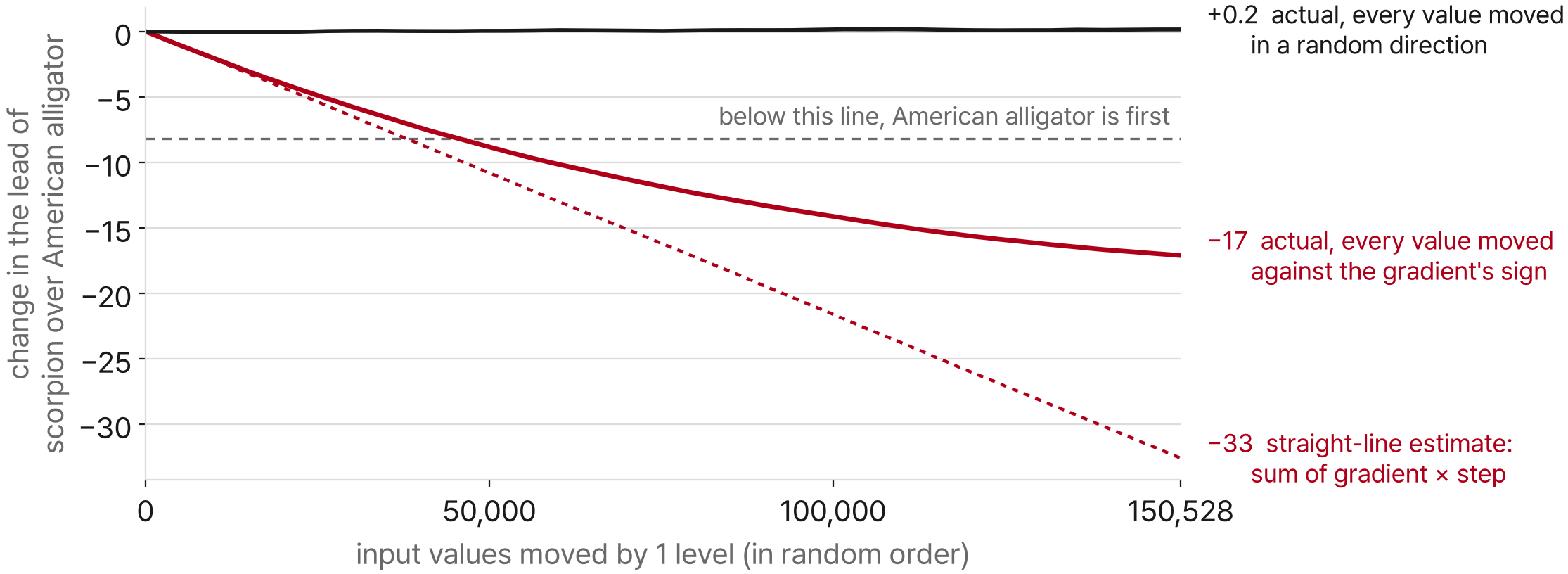
toaster, changed photo



score 31.2

# Many Small Changes Add Up

Scorpion photo from the demo: 3 × 224 × 224 = 150,528 input values



# Coming Up

- Assignment 1 due Friday, October 16, 22:00